FUNDAMENTALS OF AI ML-VITYARTHI PROJECT REPORT (BUILD YOUR OWN PROJECT)

NAME : PRANJAL BEHERA
REGISTRATION NUMBER : 25BCE10688
BRANCH- BTECH CSE CORE
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- ***UNDERSTANDING THE PROBLEM***

  With the rapid growth of digital communication, users receive a large number of unwanted messages known as **spam**. These messages often include advertisements, phishing attempts, or fraudulent links. Manually identifying such messages is time-consuming and unreliable.

  Thus keeping this problem in mind,I build a project which is a **"SPAM DETECTION TOOL"** which focuses on **"How can we automatically detect whether a messege is spam or not using AI and Machine Learning?".**

- ***WHY IT MATTERS?***

  Spam detection is important because:

  - **Security Risk:** Spam messages may contain phishing links or scams
  - **Time Waste:** Users spend time filtering irrelevant messages
  - **Productivity Loss:** Important messages may get lost among spam
  - **Privacy Threat:** Some spam tries to steal personal information

  Manual filtering is completely impractical at modern email volumes, which is why automated detection is essential. This makes spam detection a real-world, high-impact AI application used in email systems like Gmail.

- ***APPROACH TO SOLVING THE PROBLEM***

  To solve this problem,I used concepts of AI and Machine learning and also learnt some concepts from my course curriculum of Fundamentals of AI ML.

  Here's a list of concepts that are being used in this project and relevant to the course:

  1. Machine Learning Approach

  - Used a **Naive Bayes classification algorithm**
  - Imported a dataset consisting of spam and ham messages from Kaggle[spam.csv]
  - Trained on this labeled dataset (spam/ham messages)

  2. Text Processing

  - Converted text into numbers using **CountVectorizer**
  - This allowed the model to understand textual data

  3. Search Algorithm

  - Added a **Linear Search algorithm** to detect common spam keywords
  - This improves speed for obvious spam messages

  4. Graphic User Interface

  - Built a simple **GUI using Tkinter**
  - Allows users to input messages and get predictions instantly

- ***System Workflow***

  The system works in the following steps:

1. User enters a message
2. System checks for spam keywords using Linear Search
3. If not detected, message is passed to ML model
4. Text is converted into numerical form
5. Model predicts spam or not spam
6. Result is displayed on screen

- ***KEY DECISIONS MADE***

  While building this project,I made few key decisions as well which are listed below:

  - **Choice of Algorithm**

    Selected **Naive Bayes** because it's

    -Simple and beginner-friendly

    -Works well for text classification

    -Fast and efficient
  - Used **CountVectorizer** as it converts text into numerical features for model training
  - **Dataset Selection**

    Used **SMS Spam Collection Dataset** from Kaggle because its well-structured and commonly used for beginners
  - **Adding the Search Algorithm**

    Implemented **Linear Search** for keyword detection as it makes system faster for obvious spam messages
  - **GUI Implementation**-Used **Tkinter** for simplicity and ease of use

- ***PROBLEMS FACED ON THE WAY***

  As a beginner, I faced several challenges:

  Understanding Machine Learning Concepts

- Initially found it difficult to understand how models learn from data
- Solved by breaking concepts into smaller steps

  Dataset Issues

- Faced errors like:
  - File not found
  - Incorrect column names
- Fixed it by ensuring both the dataset file and python program are in the same folder and also modified the dataset format.

  Converting Text to Numbers

- Confusing at first how text becomes numerical
- Learned about vectorization techniques

  Debugging Errors

- Errors while running code particularly in file paths and functions
- Solved through trial and step-by-step debugging

  GUI Integration

- Connecting ML model with Tkinter was tricky initially

- ***KEY LEARNINGS***

  This project helped me understand:

  Technical Learnings

  - Basics of Machine Learning pipeline
  - Difference between training and testing data
  - How Naive Bayes works for classification
  - Importance of feature extraction
  - How to use Python libraries like Pandas and Scikit-learn

  Practical Skills

  - Writing structured Python code
  - Debugging real errors
  - Building a working application
  - Creating GUI for user interaction

  Conceptual Understanding

  - Difference between AI, ML, and traditional programming
  - How systems can learn from data instead of rules

- ***THE RESULTS AND THE FINAL OUTCOME***

  - Successfully built a working spam detection system
  - Achieved accuracy of approximately 90–97%
  - System correctly identifies spam and non-spam messages
  - GUI makes it user-friendly and interactive

  The final output attached at last*

- ***LIMITATIONS***

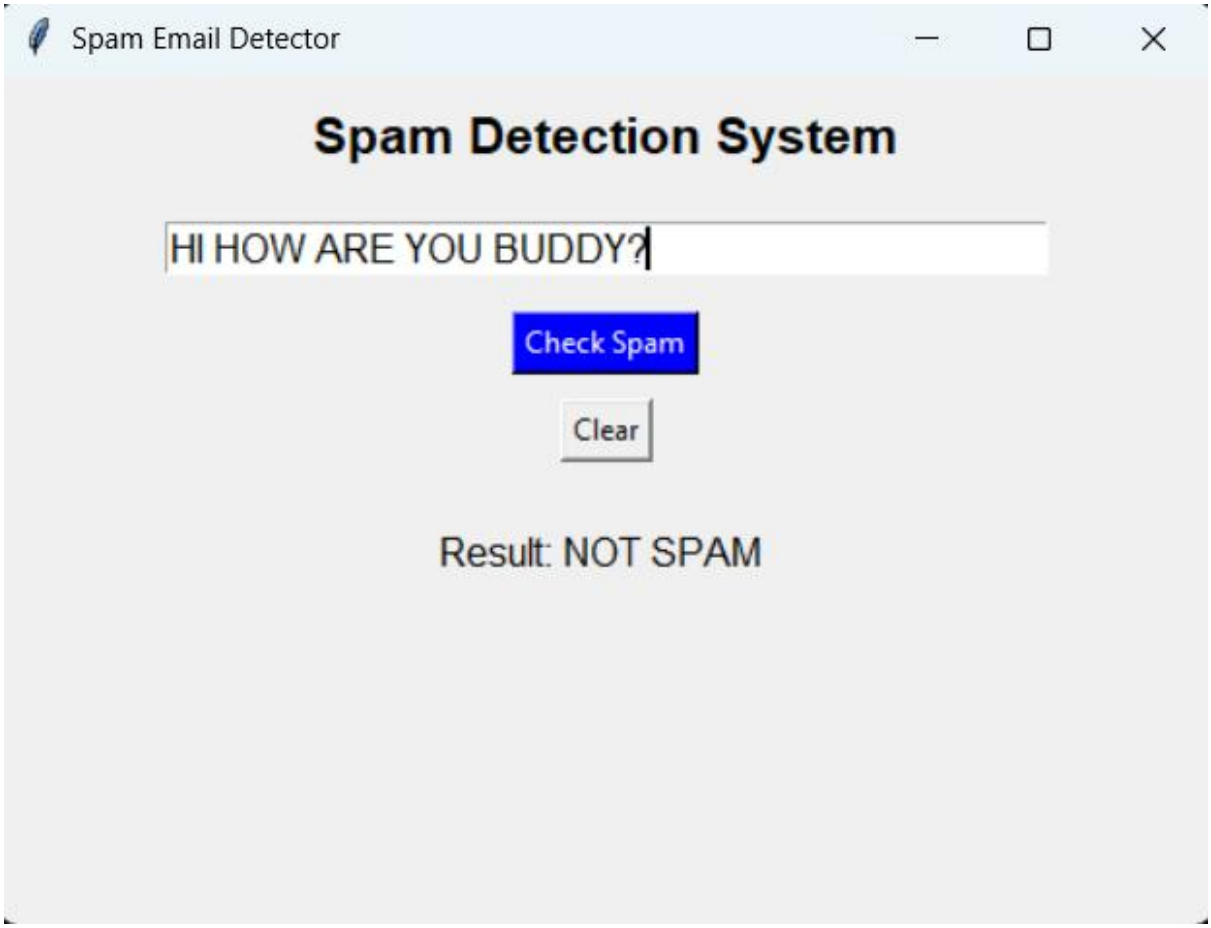
  -Model depends on dataset quality

  -May fail on complex or new spam patterns

  -Keyword search is basic and not context-aware

- ***CONCLUSION***

  This project demonstrates how Machine Learning can be applied to solve real-world problems like spam detection. By combining a Naive Bayes model with basic search algorithms and a user-friendly interface, the system efficiently classifies messages. As a beginner, this project provided valuable hands-on experience in AI/ML and helped build a strong foundation for future projects.

SAMPLE OUTPUTS- SPAM MESSAGES

NOT SPAM MESSAGES-SAFE TO GO!